PROJECT TEST PLAN & VALIDATION REPORT

Unit Tests, Verification Results, and Performance Latency Analysis

1. Testing Strategy

The testing strategy involved two phases: Unit Testing of separate modules using mock data, and End-to-End Testing of the pipeline (Audio Input → Transcription → Semantic Similarity → Feature Extraction → Grading → PDF generation).

2. Performance Benchmark Metrics

We executed an automated benchmark script (`benchmark_performance.py`) inside the project environment. Results show severe latency reduction due to caching:

• **Whisper Transcription**: Decoded a 33-second WAV recording in `5.11s` during initial model load, and in `1.26s` on subsequent runs.

• **Sentence-BERT Semantic Matching**: Took `11.89s` during the cold cache run (due to model loading), and dropped to **`0.0288s`** during warm cache runs.

• **Librosa Acoustic Features**: Extracted RMS volume, duration, and pause statistics in `1.29s` for a long audio file, and in `0.004s` for a short file.

3. Robustness & Edge Cases Tested

• **Silent Recordings**: Successfully caught by the system, outputting 0 words and showing a warning to prevent database contamination.

• **Long Audio Inputs**: Handled cleanly by Librosa resampling and Whisper decoding without memory leaks or execution timeouts.

• **Encoding Failures**: Emoji character safety was verified by checking that the PDF report generates without crashes.